Chaitanya Kapoor

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(Nationality: Citizen of the United States of America)

Research Interests

I develop mathematically rigorous, quantitative tools grounded in optimal transport theory, using high-performance computing to analyze complex, high-dimensional systems at scale. My research applies this toolkit to biological and artificial neural systems. I am keen to bring these techniques to domains that demand the same quantitative rigor and expertise.

Education

University of California, San Diego **Sep. 2025 – Present**
PhD Student (Cognitive Science) *La Jolla, CA*

University of California, San Diego **Aug. 2024 – Aug. 2025**
Post-Baccalaureate Student (NeuroML Group) *La Jolla, CA*

Advisor: Meenakshi Khosla

Massachusetts Institute of Technology **Jan. 2024 – July 2024**
Visiting Student (Senseable Intelligence Group) *Cambridge, MA*

Thesis: LUMIN: Light-sheet Microscopy Analysis Unified with Distributed and Domain-Randomized Generative Models

Advisor: Satrajit Ghosh

Birla Institute of Technology and Science, Pilani **Nov. 2020 – July 2024**
B.E. (Hons.) in Electrical and Electronics engineering *Pilani, India*

Selected Publications

Topographic Soft Matching Distance (TSMD): A Framework for Comparing the Spatial Organization of Neural Tuning
C. Kapoor, *N. A. Murty, M. Khosla*
Workshop on Symmetry and Geometry in Neural Representations (NeurIPS Workshop), 2026 - **paper**

Partial Soft-Matching For Neural Representational Comparison
C. Kapoor, *A. H. Williams, M. Khosla*
International Conference on Learning Representations (ICLR), 2026 - **paper, code**

Bridging Critical Gaps in Convergent Learning: How Representational Alignment Evolves Across Layers, Training, and Distribution Shifts
C. Kapoor, *S. Srivastava, M. Khosla*
Neural Information Processing Systems (NeurIPS), 2025 - **paper**, **code**, **CCN Extended Abstract**

Multiplexed Expansion Revealing for Imaging Multiprotein Nanostructures in Healthy and Diseased Brain
J. Kang, M. Schroeder, Y. Lee, ***C. Kapoor****, E. Yu, T. B. Tarr, K. Titterton, M. Zeng, D. Park, E. Niederst, D. Wei, E. S. Boyden*
Nature Communications, 2024 - **paper**, **code**, **news story**

RnR-ExM: Robust Non-Rigid Registration Challenge for Expansion Microscopy Volumes
E. Besier, R. Zhang, Y. Bando, Y. Quéméner, ***C. Kapoor****, M. Alawi, M. Hoffman, A. Dalca, A. Casamitjana, I. Arganda-Carreras, E. S. Boyden, H. Pfister, D. Wei*
IEEE International Symposium on Biomedical Imaging (IEEE ISBI), 2023 - **code**, **website**

Research Experience

Computational Neurobiology Laboratory **Jan. 2026 – Present**
Supervisor: Prof. Terrence J. Sejnowski *La Jolla, CA*

- Investigating traveling-wave-driven synaptic plasticity in large-scale sparse spiking networks. We propose a mechanistic link between cortical wave dynamics and Hebbian learning via temporally ordered spike sequences, STDP, and biophysical mechanisms including rebound inhibition and backpropagating action potentials.

- Developing a high-performance `C++20` spiking network simulator handling millions of nodes with sparse connectivity, sub-millisecond timing precision, and structured binary I/O (custom spike-time formats at network scale).

NeuroML Lab **Aug. 2024 – Present**
Supervisor: Prof. Meenakshi Khosla *La Jolla, CA*

- Leading the development of an optimal-transport-based topographic similarity metric that jointly accounts for neural tuning similarity and cortical geometry over different metric spaces; validated on datasets spanning salt-and-pepper mouse V1 and columnar primate cortex.
- Led the development and design of new optimal transport-based representational similarity metrics allowing for matching and detection of single-neuron tuning outliers across biological and artificial neural networks. This work has been published at ICLR 2026.
- Led a large-scale study spanning CNNs, ViTs, and LLMs to characterize how internal representations converge across architectures; found that simple geometric invariances drive alignment, cross-network convergence emerges far earlier than previously thought, and alignment robustness degrades proportionally with distributional shift. This work is published at NeurIPS 2025.

Talmo Lab **May 2023 – Aug. 2024**
Supervisor: Dr. Talmo Pereira *Salk Institute, La Jolla, CA*

- Worked on the development of methods for pose estimation and tracking to quantify animal behavior through robust tracking of anatomical landmarks in 3D.
- Designed algorithms utilizing projective geometry to create inputs for a CNN that leverages geometric reasoning.

Senseable Intelligence Group **Jan. 2024 – July 2024**
Supervisor: Prof. Satrajit Ghosh *Massachusetts Institute of Technology, MA*

- Developed a distributed framework for large-scale ($\approx$ 1TiB) image segmentation for light-sheet microscopy images.
- Designed a domain-randomized generative model, using spherical harmonics to synthesize cortical sections of *ex-vivo* human brains to enable zero-shot segmentation.
- Worked on designing few-shot, *spatially* guided semantic segmentation algorithms for MRI segmentation.
- Proposed the use of a topologically aware Graph Neural Network which is capable of encoding spatial and anatomical constraints.

Synthetic Neurobiology Group **Feb. 2022 – Jan. 2024**
Supervisors: Prof. Edward S. Boyden, Prof. Donglai Wei *Massachusetts Institute of Technology, MA*

- Worked on the development of a *generalist* 3D segmentation model for Expansion Microscopy (ExM) volumes for various animal species.
- Proposed a human-in-the-loop feedback learning mechanism, built on top of an existing deep learning framework, NucMM.
- Worked on developing a new joint-intensity and point-based, high throughput image registration algorithms having nanoscale precision ($\mathbf{10-40}$ nm) for Multiplexed Expansion Revealing **(multiExR)**. This work has been published at Nature Communications.

Invited Talks and Tutorials

03/26 **Comparative Analysis of Neural Population Codes**, *COSYNE 2026*
02/24 **Automating behavior quantification using deep learning**, *COSYNE 2024*
06/23 **SIS Symposium, Harvard SEAS,** *Seeing Beyond the Camera*

Honors & Awards

10/25 **NeurIPS 2025**: Student Travel (Scholar) Award
04/24 **NeuroMatch Academy**: Computational Neuroscience Summer School
02/24 **COSYNE 2024:** Undergraduate Travel Grant Award
04/23 **OxML 2023:** ML $\times$ Health Summer School, Oxford Mathematical Institute

Students Mentored

- **Pratham Yashwante**: Masters Student at UC San Diego
- **Chanyoung Park**: Undergraduate Student at UC San Diego

Technical Skills

Languages: Python, C, C++, MATLAB, Unix Shell Scripting , LaTeX
Technologies/Frameworks: Keras, Tensorflow, Numpy, PyTorch, Git, OpenCV

References

Meenakshi Khosla: Assistant Professor of Cognitive Science and Computer Science and Engineering, UC San Diego. (Contact: mkhosla@ucsd.edu)
Edward S. Boyden: Y. Eva Tan Professor in Neurotechnology, MIT. (Contact: eboyden@mit.edu)
Satrajit S. Ghosh: Director of Open Data in Neuroscience Initiative and Senior Research Scientist, MIT. (Contact: satra@mit.edu)
Talmo D. Pereira: Salk Fellow and Principal Investigator, Salk Institute for Biological Studies. (Contact: talmo@salk.edu)
Donglai Wei: Assistant Professor of Computer Science, Boston College. (Contact: donglai.wei@bc.edu)